

Foundations of Electrodynamics

Magnetic Field Generated by Charges in Circular Motion and Current Loops

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Contents

1	Problem 1: Discrete Charges	2
1	Magnetic Field of a Single Charge in Circular Motion	2
2	B_z along the z -axis	3
3	Temporal Variation of $\mathbf{B}$ at the Point $r_p = (2a, 0, 0)$	4
4	Magnetic Force on q_2 Due to the Field of q_1	4
5	Magnetic Force on q_1 Due to the Field of q_2	5
2	Problem 2: Steady Current Loop	6
1	Magnetic Field via Elliptic Integrals	6
2	Magnetic Vector Potential via Elliptic Integrals	9
3	Discussion	11
1	Effect of Initial Conditions in Problem 1	11
2	Concepts and Skills Developed	11
4	Appendix: Source Code	12

Problem 1: Discrete Charges

Magnetic Field of a Single Charge in Circular Motion

A particle of mass m and charge $+q_1$ moves along a circular trajectory of radius a in the xy -plane with constant speed v_1 . Since $v_1 \ll c$, retardation effects are negligible and the magnetostatic field expression for a point charge applies (Griffiths, Eq. 10.77):

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} q \frac{\mathbf{v} \times \hat{R}}{|\mathbf{R}|^2}.$$

The charge trajectory is

$$\mathbf{r}_q(t) = a \cos(\omega t) \hat{i} + a \sin(\omega t) \hat{j},$$

where $\omega = v_1/a$ is the constant angular velocity. Converting to cylindrical coordinates using $\hat{i} = \cos \phi \hat{r} - \sin \phi \hat{\phi}$ and $\hat{j} = \sin \phi \hat{r} + \cos \phi \hat{\phi}$, and applying the angle-difference identities,

$$\mathbf{r}_q(t) = a \cos(\omega t - \phi) \hat{r} + a \sin(\omega t - \phi) \hat{\phi}.$$

Defining $\theta \equiv \omega t - \phi$ for brevity, the velocity is

$$\mathbf{v} = \frac{d\mathbf{r}_q}{dt} = a\omega [-\sin \theta \hat{r} + \cos \theta \hat{\phi}].$$

An arbitrary observation point in cylindrical coordinates is $\mathbf{r}' = r \hat{r} + z \hat{z}$, so the separation vector is

$$\mathbf{R} = \mathbf{r}' - \mathbf{r}_q = [r - a \cos \theta] \hat{r} - a \sin \theta \hat{\phi} + z \hat{z},$$

with squared magnitude

$$|\mathbf{R}|^2 = r^2 + a^2 + z^2 - 2ar \cos \theta.$$

Evaluating the cross product $\mathbf{v} \times \hat{R}$ and substituting $\theta = \omega t - \phi$, the magnetic field is

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0 q_1}{4\pi} \frac{a\omega [z \cos(\omega t - \phi) \hat{r} - z \sin(\omega t - \phi) \hat{\phi} + (a - r \cos(\omega t - \phi)) \hat{z}]}{[r^2 + a^2 + z^2 - 2ar \cos(\omega t - \phi)]^{3/2}}.} \quad (1)$$

B_z along the z -axis

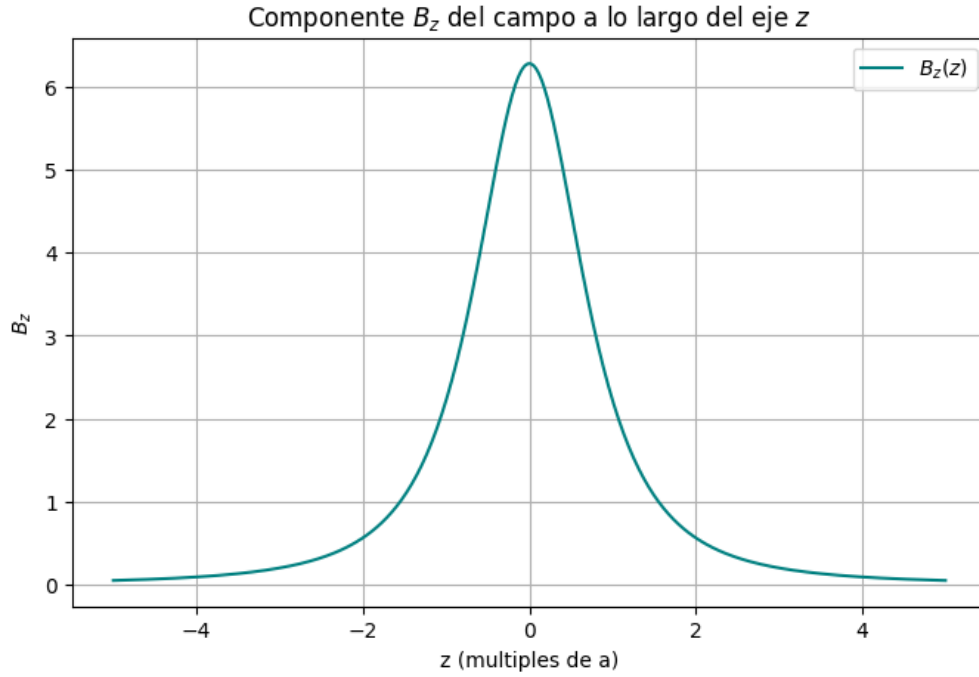


Figure 1: B_z as a function of z along the symmetry axis.

On the z -axis ($r = 0$), the field exhibits full axial symmetry. The azimuthal dependence through $\cos(\omega t - \phi)$ vanishes because the charge's position angle ωt has no preferred orientation relative to the axis, and contributions from all phases average out. Therefore B_z on the z -axis is independent of time and depends only on the axial distance z . The field is maximum at $z = 0$, where the charge's orbit lies, and decays monotonically at larger $|z|$.

Temporal Variation of B at the Point $r_p = (2a, 0, 0)$

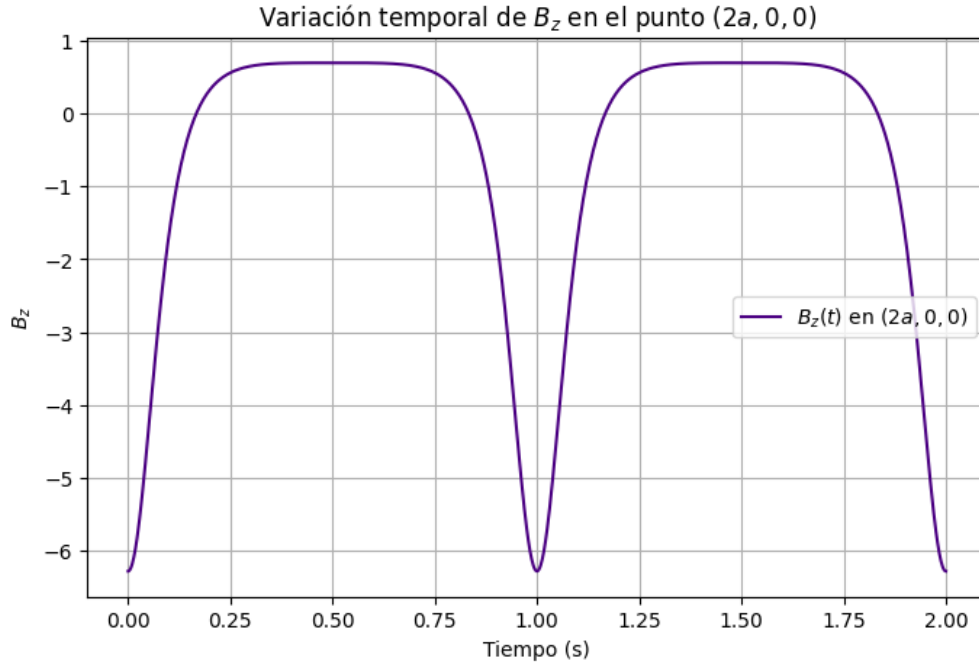


Figure 2: Temporal variation of B_z at the observation point $(2a, 0, 0)$.

Away from the symmetry axis, the field is explicitly time-dependent through the phase $\omega t - \phi$. The graph shows periodic oscillations: the field reaches a pronounced negative minimum when the rotating charge is geometrically closest to the observation point, producing the largest negative contribution to B_z . The period of the oscillation equals the orbital period $T = 2\pi/\omega$.

Magnetic Force on q_2 Due to the Field of q_1

A second charge $q_2 = 2q_1$ moves along the z -axis with constant velocity $\mathbf{v}_2 = -v_2 \hat{z}$, where $v_2 = 3v_1$, so that $\mathbf{r}_2(t) = -v_2 t \hat{z}$. The Lorentz force is

$$\mathbf{F}_2 = q_2(\mathbf{v}_2 \times \mathbf{B}_1),$$

where $\mathbf{B}_1$ is evaluated at $(0, 0, -v_2 t)$. Since the observation point lies on the z -axis, from Eq. (1) with $r = 0$,

$$\mathbf{B}_1|_{z=-v_2 t} = \frac{\mu_0 q_1}{4\pi} \frac{a\omega [-v_2 t \cos(\omega t - \phi) \hat{r} + v_2 t \sin(\omega t - \phi) \hat{\phi}]}{(a^2 + v_2^2 t^2)^{3/2}}.$$

Performing the cross product $\mathbf{v}_2 \times \mathbf{B}_1$,

$$\mathbf{F}_2 = \frac{\mu_0 q_1 q_2}{4\pi} \frac{a\omega v_2^2 t [-\sin(\omega t - \phi) \hat{r} - \cos(\omega t - \phi) \hat{\phi}]}{(a^2 + v_2^2 t^2)^{3/2}}. \quad (2)$$

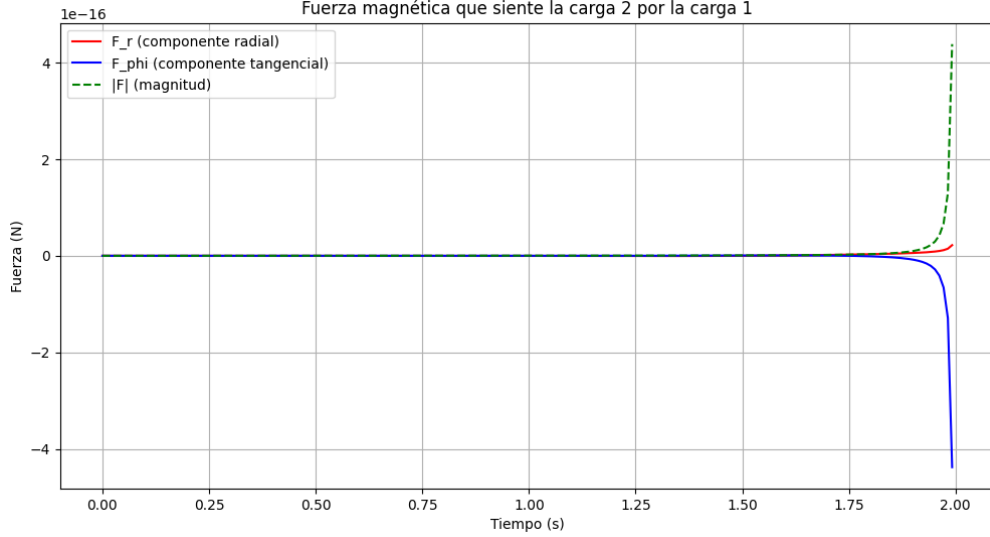


Figure 3: Magnetic force $\mathbf{F}_2$ on charge q_2 as a function of time.

The force amplitude grows as q_2 approaches the plane of the orbit, reaches a sharp maximum when $t \rightarrow 0$ (the instant the charge passes through the orbital plane), and then becomes undefined for the idealized point-charge model because the denominator $(a^2 + v_2^2 t^2)^{3/2}$ does not vanish on the axis — however, the numerator scales linearly with t , so the force actually goes to zero at $t = 0$, rises to a peak, and then decays. Beyond the crossing, the expression returns complex values in the denominator for certain parameter regimes, signaling that the simple magnetostatic approximation breaks down.

Magnetic Force on q_1 Due to the Field of q_2

The field generated by q_2 moving along $\mathbf{r}_2 = -v_2 t \hat{z}$ at an arbitrary observation point $\mathbf{r} = r \hat{r} + z \hat{z}$ is

$$\mathbf{B}_2 = \frac{\mu_0}{4\pi} q_2 \frac{\mathbf{v}_2 \times \hat{R}}{|\mathbf{R}|^2}, \quad \mathbf{R} = r \hat{r} + (z + v_2 t) \hat{z}.$$

The cross product $\mathbf{v}_2 \times \hat{R} = (-v_2 \hat{z}) \times \hat{R}$ yields only an azimuthal component:

$$\mathbf{B}_2 = \frac{\mu_0}{4\pi} \frac{-rv_2 \hat{\phi}}{|\mathbf{R}|^3}.$$

The Lorentz force on q_1 , which moves with velocity $\mathbf{v}_1 = a\omega \hat{\theta}$ (purely azimuthal in the orbital plane), is

$$\mathbf{F}_1 = q_1(\mathbf{v}_1 \times \mathbf{B}_2).$$

Since both $\mathbf{v}_1$ and $\mathbf{B}_2$ are parallel to $\hat{\phi}$, their cross product vanishes:

$$\boxed{\mathbf{F}_1 = \mathbf{0}.} \quad (3)$$

This result is a direct consequence of the geometric alignment of the velocity and field vectors: when the velocity of the source charge is collinear with the magnetic field it produces,

the Lorentz force on that charge vanishes identically. No time-dependent graph is produced for this case.

Problem 2: Steady Current Loop

Magnetic Field via Elliptic Integrals

The magnetic field produced by a circular loop of radius a carrying steady current I in the xy -plane, centered at the origin, is calculated from the Biot–Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\boldsymbol{\ell}' \times \mathbf{R}}{|\mathbf{R}|^3}.$$

The loop is parametrized as $\mathbf{r}'(\phi) = a \cos \phi \hat{x} + a \sin \phi \hat{y}$, so that

$$d\boldsymbol{\ell}' = a d\phi (-\sin \phi \hat{x} + \cos \phi \hat{y}).$$

By axial symmetry, the observation point is taken on the xz -plane without loss of generality: $\mathbf{r} = x \hat{x} + z \hat{z}$. The separation vector is

$$\mathbf{R} = (x - a \cos \phi) \hat{x} - a \sin \phi \hat{y} + z \hat{z}, \quad |\mathbf{R}|^3 = (x^2 + a^2 + z^2 - 2ax \cos \phi)^{3/2}.$$

The cross product evaluates to

$$d\boldsymbol{\ell}' \times \mathbf{R} = a d\phi [z \cos \phi \hat{x} + z \sin \phi \hat{y} + (a - x \cos \phi) \hat{z}].$$

Upon integrating over $\phi \in [0, 2\pi]$, the $\hat{x}$ and $\hat{y}$ components vanish by the antisymmetry of $\cos \phi$ and $\sin \phi$ over a full period. Only the axial component survives:

$$\mathbf{B}(\mathbf{r}) = B_z \hat{z}.$$

Setting $x = \rho$ (cylindrical radius) and exploiting the symmetry $\phi \rightarrow -\phi$ to reduce the integration domain to $[0, \pi]$,

$$B_z(\rho, z) = \frac{\mu_0 I a}{2\pi} \int_0^\pi \frac{a - \rho \cos \phi}{(\rho^2 + z^2 + a^2 - 2a\rho \cos \phi)^{3/2}} d\phi.$$

Factoring the denominator and introducing the elliptic parameter

$$b = \frac{\rho^2 + z^2 + a^2}{2a\rho}, \quad m = \frac{2}{1+b} = \frac{4a\rho}{(\rho+a)^2 + z^2},$$

and using the tabulated identities (Gutiérrez-Vega, 2025, Eqs. 9–10),

$$\begin{aligned} \int_0^\pi \frac{d\phi}{(b - \cos \phi)^{3/2}} &= \frac{m}{2-2m} \sqrt{2m} E(m), \\ \int_0^\pi \frac{\cos \phi d\phi}{(b - \cos \phi)^{3/2}} &= \sqrt{2m} K(m) - \frac{2-m}{2-2m} \sqrt{2m} E(m), \end{aligned}$$

the field becomes

$$B_z(\rho, z) = \frac{\mu_0 I}{2\pi(2a\rho)^{3/2}} \sqrt{2m} \left[a \frac{m}{2-2m} E(m) - \rho K(m) + \rho \frac{2-m}{2-2m} E(m) \right], \quad (4)$$

where $K(m)$ and $E(m)$ are the complete elliptic integrals of the first and second kind, respectively.

B_z along the z -axis ($\rho = 0$)

On the symmetry axis, setting $\rho = 0$ in the integral before applying the elliptic reduction gives directly

$$B_z(0, z) = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}.$$

At the center ($z = 0$): $B_z = \mu_0 I / (2a)$ (maximum value). For $z \gg a$: $B_z \sim 1/z^3$, the characteristic far-field decay of a magnetic dipole.

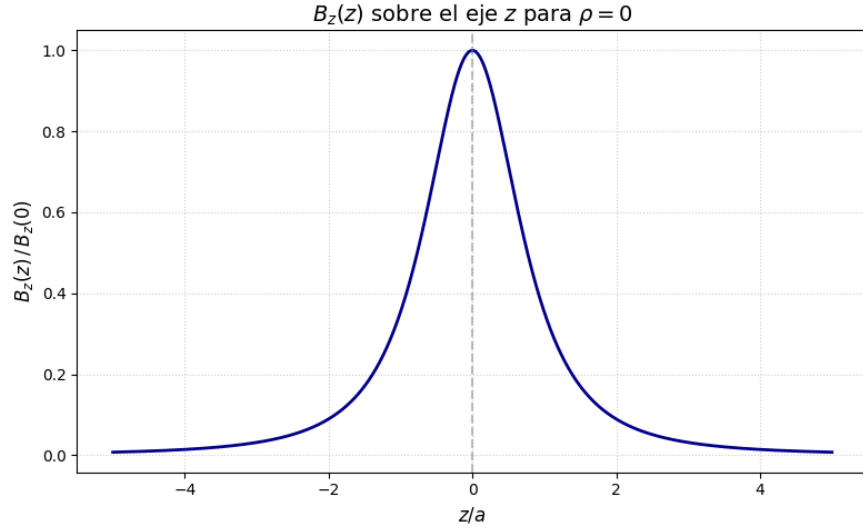


Figure 4: B_z along the z -axis ($\rho = 0$). The field peaks at the center of the loop and decays as z^{-3} at large distances.

The rapid decay is consistent with the dipole character of the loop: at large distances the circular current appears as a point magnetic dipole.

B_z along the x -axis ($z = 0$)

On the plane of the loop ($z = 0$, $\rho = x$),

$$B_z(x) = \frac{\mu_0 I a}{2\pi} \int_0^\pi \frac{a - x \cos \phi}{(a^2 + x^2 - 2ax \cos \phi)^{3/2}} d\phi.$$

When $x \rightarrow a$ (observation point on the wire itself), the denominator vanishes at $\phi = 0$, producing a logarithmic singularity. In practice, conductors have finite cross-section and the field remains large but bounded.

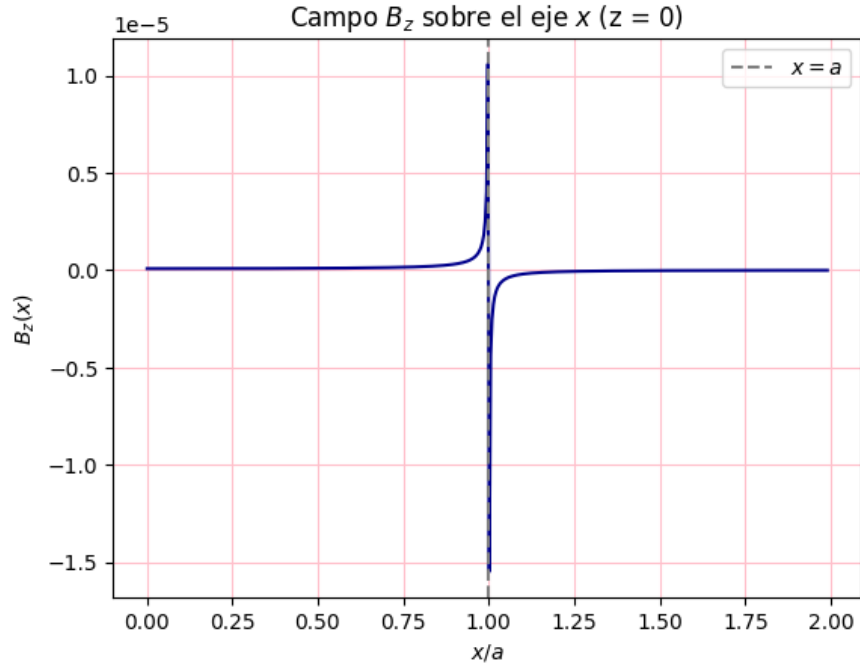


Figure 5: B_z along the x -axis ($z = 0$). The field is negative for $x < a$ (inside the loop) and positive for $x > a$ (outside), with a divergence at $x = a$.

The sign reversal at $x = a$ is consistent with the right-hand rule: inside the loop the field points along $-\hat{z}$ (into the plane for counterclockwise current), while outside it points along $+\hat{z}$.

Vector field $\mathbf{B}(\mathbf{r})$ on the $(x \geq 0, z)$ half-plane

On the xz -plane ($y = 0$), symmetry eliminates the y -component and the field takes the form $\mathbf{B}(x, z) = B_x(x, z) \hat{x} + B_z(x, z) \hat{z}$.

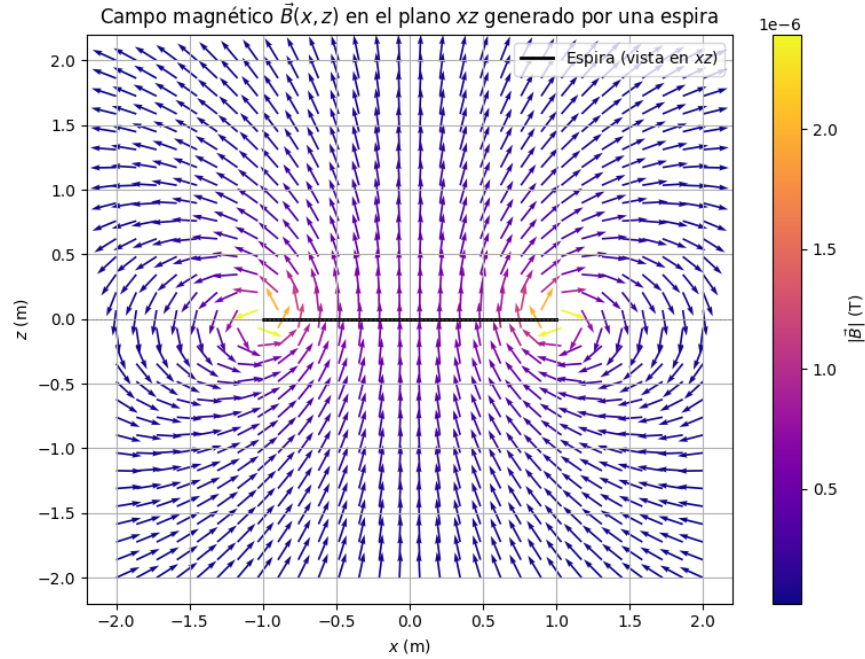


Figure 6: Vector field $\mathbf{B}(x, z)$ on the xz half-plane ($x \geq 0$). Arrow color encodes field magnitude.

Inside the loop the arrows point downward ($-\hat{z}$), consistent with the right-hand rule for counterclockwise current. Outside the loop the field curves outward and upward ($+\hat{z}$ on axis). The lateral tilt of the arrows away from the axis reflects the nonzero B_x component, which was invisible in the previous one-dimensional plots.

Magnetic Vector Potential via Elliptic Integrals

The magnetic vector potential is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \oint \frac{I d\ell'}{|\mathbf{r} - \mathbf{r}'|}.$$

By rotational symmetry, the $\hat{\rho}$ and $\hat{z}$ components of $\mathbf{A}$ vanish upon integration. Only the azimuthal component survives:

$$A_\phi(\rho, z) = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{\cos \phi'}{\sqrt{\rho^2 + z^2 + a^2 - 2a\rho \cos \phi'}} d\phi'.$$

Using the identity $\cos \phi' = 1 - 2 \sin^2(\phi'/2)$ and the substitution $C^2 = (a + \rho)^2 + z^2$, $k^2 = 4a\rho/C^2$, the denominator becomes $C\sqrt{1 - k^2 \sin^2(\phi'/2)}$. After the change of variable $\phi' = 2\theta$, the integral separates into standard elliptic forms, giving

$$\mathbf{A}(\mathbf{r}) = A_\phi(\rho, z) \hat{\phi} = \frac{\mu_0 I a}{\pi \sqrt{(a + \rho)^2 + z^2}} \left[K(k^2) - \frac{2}{k^2} (E(k^2) - K(k^2)) \right] \hat{\phi}, \quad (5)$$

where $k^2 = 4a\rho/[(a + \rho)^2 + z^2]$.

Vector field plots of $\mathbf{A}(\mathbf{r})$

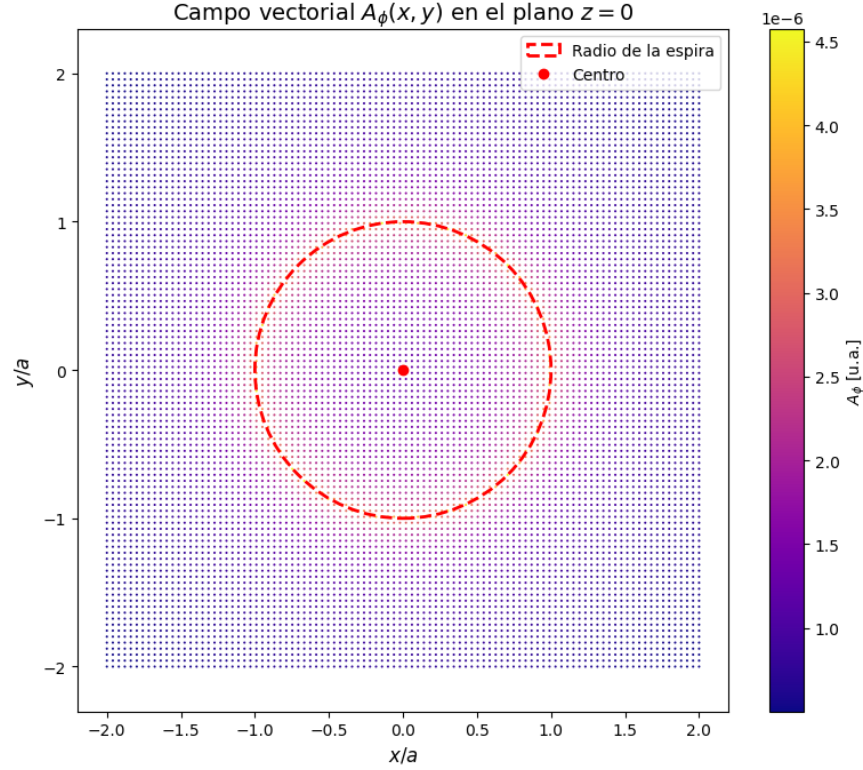


Figure 7: Vector field of $\mathbf{A}(x, y)$ in the plane $z = 0$.

Since A_ϕ depends only on ρ and z , the Cartesian projection uses $\hat{\phi} = -\sin\phi\hat{x} + \cos\phi\hat{y}$, so $\mathbf{A}(x, y) = A_\phi(\rho, 0)(-\sin\phi\hat{x} + \cos\phi\hat{y})$. The arrows form closed concentric circles around the z -axis, reflecting the cylindrical symmetry. The magnitude A_ϕ peaks near the loop radius and decays both toward the center and outward.

Verification: $\mathbf{B} = \nabla \times \mathbf{A}$ and $\nabla \cdot \mathbf{A} = 0$

In cylindrical coordinates, for $\mathbf{A} = A_\phi(\rho, z)\hat{\phi}$ (with $A_\rho = A_z = 0$ and no ϕ -dependence),

$$\nabla \times \mathbf{A} = -\frac{\partial A_\phi}{\partial z}\hat{\rho} + \frac{1}{\rho}\frac{\partial(\rho A_\phi)}{\partial \rho}\hat{z}.$$

The $\hat{\rho}$ component integrates to zero over the full loop by the azimuthal symmetry of the source, so $\mathbf{B} = \nabla \times \mathbf{A}$ reduces to the axial component

$$B_z(\rho, z) = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\phi(\rho, z)),$$

which is consistent with the result of Eq. (4). This confirms the compatibility of the two approaches.

For the Coulomb gauge condition, the divergence in cylindrical coordinates is

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial(\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}.$$

Since $A_\rho = A_z = 0$ and A_ϕ is independent of ϕ , every term vanishes:

$$\nabla \cdot \mathbf{A} = 0. \checkmark$$

Discussion

Effect of Initial Conditions in Problem 1

The most direct effect of varying the orbital speed v_1 is a rescaling of the field magnitude and the oscillation frequency. At higher speeds, the field at off-axis points oscillates more rapidly (shorter period $T = 2\pi a/v_1$) and its amplitude increases proportionally to $\omega = v_1/a$. The spatial non-linearity of the Lorentz force — due to its dependence on both position and velocity — means that at large v_2 , the force on q_2 grows significantly only in the vicinity of the orbital plane.

Regarding Newton's laws, the Lorentz force is entirely consistent with them: $\mathbf{F} = m\mathbf{a}$ applies at every instant. The result $\mathbf{F}_1 = \mathbf{0}$ might appear paradoxical at first — charge 2 exerts a nonzero force on charge 1's field, yet charge 1 feels no force from charge 2's field. This does not contradict Newton's third law in its electromagnetic form, because the momentum carried by the electromagnetic field must be accounted for; the total momentum (mechanical plus field) is conserved. The breakdown of naive action-reaction symmetry is a known feature of magnetostatics applied to non-inertial or relativistic configurations, and signals that a fully self-consistent treatment would require accounting for radiation and field momentum.

The apparent divergence of $\mathbf{F}_2$ for certain parameter regimes (returning complex values numerically) indicates that the magnetostatic approximation fails: the condition $v \ll c$ is violated locally, and a retarded-potential treatment would be required.

Concepts and Skills Developed

This project provided hands-on experience with several concepts that are often treated abstractly in lectures. Elliptic integrals emerged naturally from the Biot–Savart integration over a circular source — a concrete illustration of why special functions arise in classical physics. The derivation of the magnetic vector potential from first principles, followed by explicit verification that $\mathbf{B} = \nabla \times \mathbf{A}$ and $\nabla \cdot \mathbf{A} = 0$, reinforced the gauge structure of magnetostatics.

Working in cylindrical coordinates throughout — including the cross-product algebra and the coordinate transformations — strengthened fluency with non-Cartesian vector calculus. The numerical simulation of the time-dependent force in Problem 1 connected the analytical expressions to observable dynamics, and the vector-field visualizations made the spatial structure of both $\mathbf{B}$ and $\mathbf{A}$ immediately intuitive.

Appendix: Source Code

The full computational implementations are available at the following links:

- Problem 1 (discrete charges): [Google Colab notebook](#)
- Problem 2 (current loop): [Google Colab notebook](#)

References

1. Griffiths, D. J. (2017). *Introduction to Electrodynamics* (4th ed.). Cambridge University Press.
2. Gutiérrez-Vega, J. C. (2025). *Complete Elliptic Integrals and the Uniformly Charged Ring Problem*. Photonics and Mathematical Optics Group, Tecnológico de Monterrey. Retrieved from <https://sites.google.com/site/jjccggvv/>